

WEEKLY COLLABORATOR DISCUSSION

Both Sides Measured

Where the wafer wins, and what it cannot hold: a 4B GPU baseline, three traces re-cut, and layout C on silicon.

MODEL	RATES	PERIOD	DATE
Qwen3-4B	CS-3 tok/s at 750 MHz as reported — clock TBD	09-07 → 09-13	2026-09-13

THIS WEEK IN THREE LINES

Decode leads 4–5×, on the one workload we cannot hold.

Layout C is closed: the kernel is coordination-bound, not FLOP-bound.

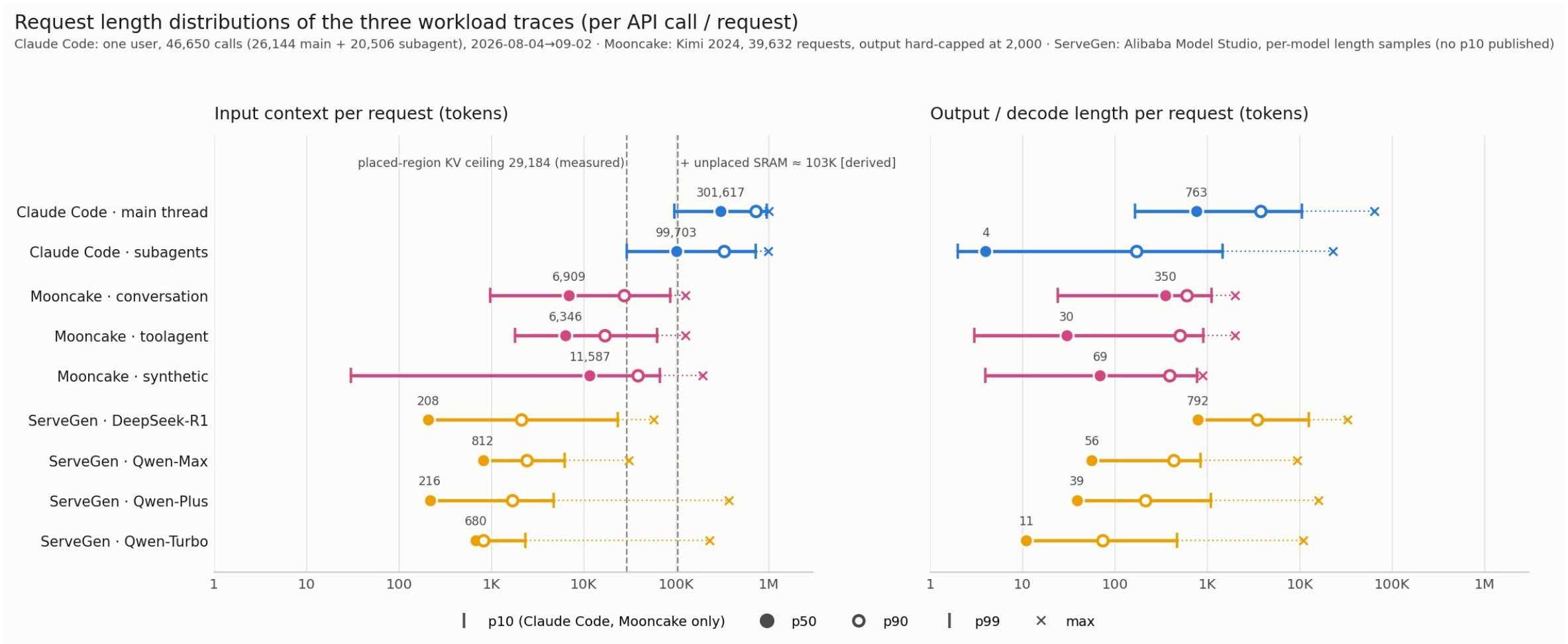
Both sides measured for the first time.

01

The workload

Three traces, re-cut per request on 2026-09-12: Claude Code (46,650 de-duplicated calls), Mooncake (39,632 requests), ServeGen (four Alibaba models).

Three worlds, not one workload



Per-request input context (left) and output length (right), log scale; p10–p99 bar, p50 filled, p90 open, × max. Dashed lines: 29,184 tokens measured placed-region ceiling, and ≈103K whole-wafer total (derived, not measured). Claude Code: one user, 26,144 main-thread + 20,506 subagent calls, 2026-08-04 → 09-02. Mooncake output is hard-capped at 2,000. ServeGen publishes no p10.

Only the agentic harness produces the case where not even one request fits

- Claude Code: huge input (p50 100–300K), medium output (a few K to ~20K per turn)
- Mooncake: input a few K, output a few hundred — p99 input 62–85K
- ServeGen R1: input a few hundred, output a few thousand — p99 input $\leq 23K$
- Every non-agentic trace has p99 input inside the measured 29,184-token ceiling
- The whole-wafer total ($\approx 103K$, derived) holds tens of such sessions — and a fraction of one Claude Code request
- So 'grow one request' vs 'park many requests' has no workload-independent answer

Subagent calls are a different shape, not a smaller copy

Tokens per call; p10 / p50 / p90 / p99 unless noted.

	MAIN THREAD (26,144 CALLS)	SUBAGENTS (20,506 CALLS)
Input context	95K / 302K / 719K / 955K	29K / 100K / 326K / 726K
First call of a session, context p50	51.5K	18.6K
Output (p10 / p50 / p90 / p99 / max)	166 / 763 / 3.8K / 10.6K / 64K	2 / 4 / 170 / 1.5K / 22.8K
Calls with output $\leq$ 5 tokens ($\leq$ 50)	0.2% (1.1%)	58% (70%)

The decode budget lives on the main thread; subagents are prefill-dominated fan-out

- Subagents are 44% of all calls, with output p50 of 4 tokens
- Main-thread output is 11.6% of that class's token traffic; subagents 1.5%
- Correction to last week: the 'median context 309K' was the main-thread number
- Including subagent calls, the per-call median is 197K — the figure used from here on

02

The GPU baseline

SGLang 0.5.19 on one H100 80GB, vanilla, every cell measured. Same model on both sides for the first time.

BASELINE · MEASURED BOTH SIDES

The wafer leads single-stream decode by 4–5× and trails prefill by 7×

Qwen3-4B. CS-3 rates as reported by the source session (750 MHz); clock convention pending.

	CS-3	H100 80GB	LEAD
Decode, bs = 1 @ 4K	~1,000 tok/s (1.0 ms)	213 tok/s (4.70 ms)	CS-3 4.7×
Decode, bs = 1 @ ~30K	653 tok/s	162 tok/s	CS-3 4.0×
Context slope	21.5 ns/token	48.9 ns/token	CS-3 2.3× shallower
Decode, batched	~2,100 tok/s	4,684 tok/s (bs 32 @ 1K)	H100 2.2×
Prefill, native	9.3–9.8K tok/s	66.5–70.4K tok/s	H100 ~7×
Max context, one request	29K	~524K	H100 18×

Our only measured advantage applies exclusively to the workload we cannot hold

- Confirmed: prefill is $\sim 7\times$ behind — finding 10 is now measured, not extrapolated
- Killed: the 'fast recompute engine for short-context multi-tenant traffic' claim
- Even the analytical thin-stage forced prefill (31–43K tok/s) is $1.6\text{--}2.2\times$ slower than the H100's measured native prefill
- There is no prefill-side wedge anywhere on the roadmap
- Median agentic turn at short context: CS-3 6.9 s vs H100 28 s
- At the real 197K context: H100 ~ 75 s, CS-3 cannot hold it at all

03

The capacity case

Every rung of the on-chip ladder against the traces, then the two ways around it: more wafers, or a link.

No rung reaches the agent harness; the workloads it reaches also fit in 80 GB of HBM

Ceilings are analytical except today's, which is measured. Trace percentiles: Claude Code p50 197K, Mooncake p99 85K, ServeGen p99 23K.

RUNG	CEILING	BASIS	CLAUDE CODE P50	MOONCAKE · SERVEGEN P99
Today	29,184	measured	no	no · yes
Fusion	~17.6K	derived	no	no · no
Split	~26K	derived	no	no · yes
Near pool	95K	derived	no	yes · yes
Far pool (every lever)	167K	derived	no — below the median	yes · yes
fp8 + far pool + chunked scoring	≈225K	derived	yes — ~55% of calls	yes · yes

CAPACITY · IN ONE SENTENCE

Capacity cannot reach the workload that wants it; the workloads it reaches do not need it.

The reopener is the score-buffer wall: fp8 storage plus chunked scoring would clear the 197K median for ~55% of agent calls — all derived.

Multi-card does not rescue capacity: it halves the one thing we lead

- Measured: 175 μ s per wafer per token at today's layout
- 29K $\rightarrow$ 197K needs $\approx$ 6 extra wafers, i.e. +1.01 ms per token (derived)
- Decode goes from 1.05 to 2.06 ms per token: 954 $\rightarrow$ 485 tok/s
- A negative-sum trade in the one dimension where the wafer is ahead

Per token, attention reads the whole KV — it cannot cross a link

Card-to-card interconnect at 1.2 Tbit/s (150 GB/s), per Le; not measured in this project.

Whole decode step on the wafer is 1.0 ms.

CONTEXT	KV	OVER THE 150 GB/S LINK	H100 FROM ITS OWN HBM
29K	4.01 GB	26.7 ms / token	1.20 ms
197K	27.1 GB	180 ms / token	8.08 ms

KV must live where the attention compute is; the link carries activations, and AFD is a weights-side solution

- Activations: 5 KB per layer, 34 ns of wire — pure latency
- $36 \text{ layers} \times 6\text{--}10 \text{ }\mu\text{s round trip} = 0.22\text{--}0.36 \text{ ms per token}$: affordable
- So attention–FFN disaggregation is viable on the real interconnect
- But it removes FFN weight traffic: worth 57.3% at 4K, only 15.0% at 197K
- A dense 4B on an 80 GB GPU is AFD's worst case — which is why NVIDIA keeps attention + KV together

04

The prefix × decode spectrum

Four configurations across eleven workload scenarios, derived from each machine's measured rates. Per-request wall time.

PD-split wins every scenario that fits: the GPU prefills, the wafer decodes

ms per request. PD-split = GPU prefill → link → CS-3 decode. Derived from measured rates; link at 0.98 μs per prefix token (unmeasured).

SCENARIO (PREFIX / DECODE)	CS-3 ALONE	PD-SPLIT	H100 ALONE	GPU-STREAMED
Qwen-Turbo p50 (680 / 11)	82	21	61	28
ServeGen R1 p50 (208 / 792)	745	726	3,654	888
Long reasoning (208 / 10,000)	9,147	9,128	46,105	11,172
R1 p90 (2,100 / 3,400)	3,462	3,273	16,020	10,290
Qwen-Max p90 (2,400 / 433)	668	453	2,078	1,472

The longer the prefix, the more the 7× prefill deficit costs CS-3 alone

ms per request. Same model as the previous slide.

SCENARIO (PREFIX / DECODE)	CS-3 ALONE	PD-SPLIT	H100 ALONE	GPU-STREAMED
Mooncake conv p50 (6,900 / 350)	1,096	482	1,834	2,850
Mooncake toolagent p90 (17,000 / 507)	2,435	1,012	3,103	9,468
Long-doc classification (27,000 / 20)	2,872	734	796	1,238
Mooncake conv p90 (27,000 / 597)	3,730	1,592	4,212	17,411

Above 29K only the H100 serves — PD-split does not extend the ceiling, because its decode side is the wafer

ms per request. The Claude Code row is a frontier-model workload a 4B never serves; it is here for completeness, not as a target.

SCENARIO (PREFIX / DECODE)	CS-3 ALONE	PD-SPLIT	H100 ALONE	GPU-STREAMED
Mooncake conv p99 (85,000 / 1,100)	no fit	no fit	12,075	97,363
Claude Code median (197,000 / 5,800)	no fit	no fit	88,215	1,158,677

Prefix length sets the margin over CS-3 alone; decode share sets the margin over the H100

- Over CS-3 alone: 1.03× at a 208-token prefix, rising to 3.9× at 27K — PD-split is the prefill fix
- Over H100 alone: 5.0× where decode is ~100% of the work, down to 1.08× at 4% — the decode advantage
- The 29K wall decides participation; above it the GPU is alone, with or without a wafer
- Short-prefix / long-decode (reasoning) is the wafer's home turf: 5× over H100 alone, PD-split barely needed
- Long-prefix / short-decode (RAG, summarisation) needs PD-split, and the H100 is competitive
- PD-split was ruled out earlier on a 3.2 GB/s H2D measurement; the real link is 62–176× that requirement

The same 167K holds zero agentic requests or 803 ServeGen requests

Capacity spent on one request's depth, or on many requests' prefixes. Derived.

CAPACITY	DEPTH: ONE AGENTIC REQUEST	MOONCAKE P50 (6.9K PREFIX)	SERVEGEN R1 P50 (208 PREFIX)
Today, 29K (measured)	14.7% of the 197K median	4 prefixes	139 prefixes
Far pool, 167K (derived)	84.8% — still short	24 prefixes	803 prefixes

Push breadth first: prefix reuse inside today's 29K, with harvesting as the mechanism

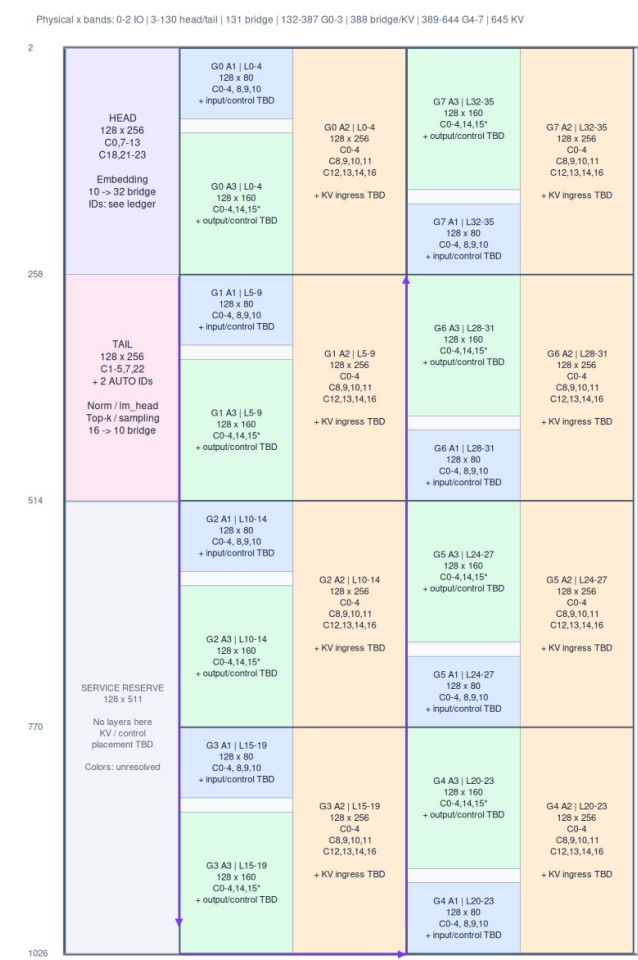
- Prefix reuse is startable inside today's 29K, with two independent justifications
- Our prefill is 7× slower, so avoiding a re-prefill is worth 7× more to us: 10.9K tokens cost the H100 156 ms and CS-3 1,147 ms
- ServeGen traffic is 3.9 : 1 prefill-to-decode by token
- Near-pool harvesting is the one rung already priced: 1.17 cycles per word, inside the ≤ 2 gate (simfab microbench)
- Then the depth-vs-breadth allocation curve
- Do not chase 167K or 225K as a target

05

Layout C

One decoder layer cut into three thin stages on a single 256×256 block. Implemented through a five-layer group, measured on CS-3, closed on 2026-09-13.

Eight groups of five layers on a 650 × 1028 footprint, with head and tail on the left

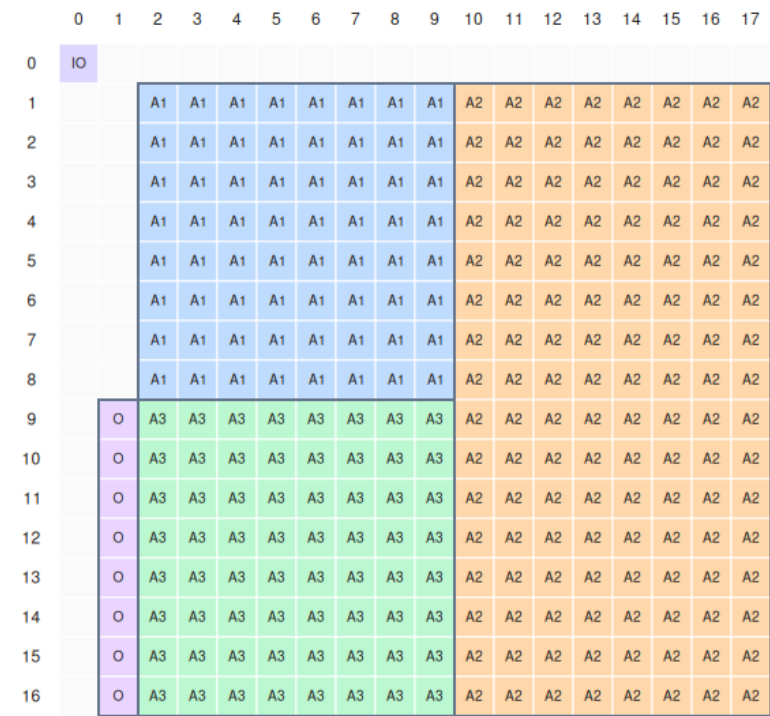


Wafer map only; the colour ledger is in the source figure. Analytical placement from a design and source audit, not a compiled full model.

A1 and A3 share the left half; A2 takes the whole right half

Layout C: tall A2 miniature

Compiled placement | 16 x 16 core | one layer | simulator validation only



Core regions (inclusive coordinates)

A1: x=2..9, y=1..8 8 x 8
A2: x=10..17, y=1..16 8 x 16
A3: x=2..9, y=9..16 8 x 8

QKV: preserve local column c

A1 diagonal PE (c,c) -> east boundary
-> A2 diagonal PE (c,c) -> entire column c.
Each column retains its own 24-value shard.
Only the current KV-owner row appends K/V.

X and Z: change row ownership

A1 row r: X[8r : 8r+8]
A2 row r: first 4; row 8+r: last 4.
Z merges those same two rows at row 8+r,
then travels west to A3 row r.

Boundary fixtures and output

Embeddings + prefill are baked into this test.
O: output collector at x=1, y=9..16.
IO: SDK output adaptor at (0,0).
No head/tail kernels or layer feedback yet.

Miniature scales the requested 256 x 256 geometry: A1/A3 each 128 x 128 on the left; A2 128 x 256 on the right.

Coordinates are SdkLayout fabric coordinates. The separate sim_cut3 figure remains a historical testbed diagram.

Compiled miniature (16 × 16 core, one layer, simulator validation) with the same shape as the full geometry: A1 128 × 80 (32 hidden per row), spare 128 × 16, A3 128 × 160 (16 per row), A2 128 × 256 (10 per row). 65,536 PEs reserved, 63,488 active.

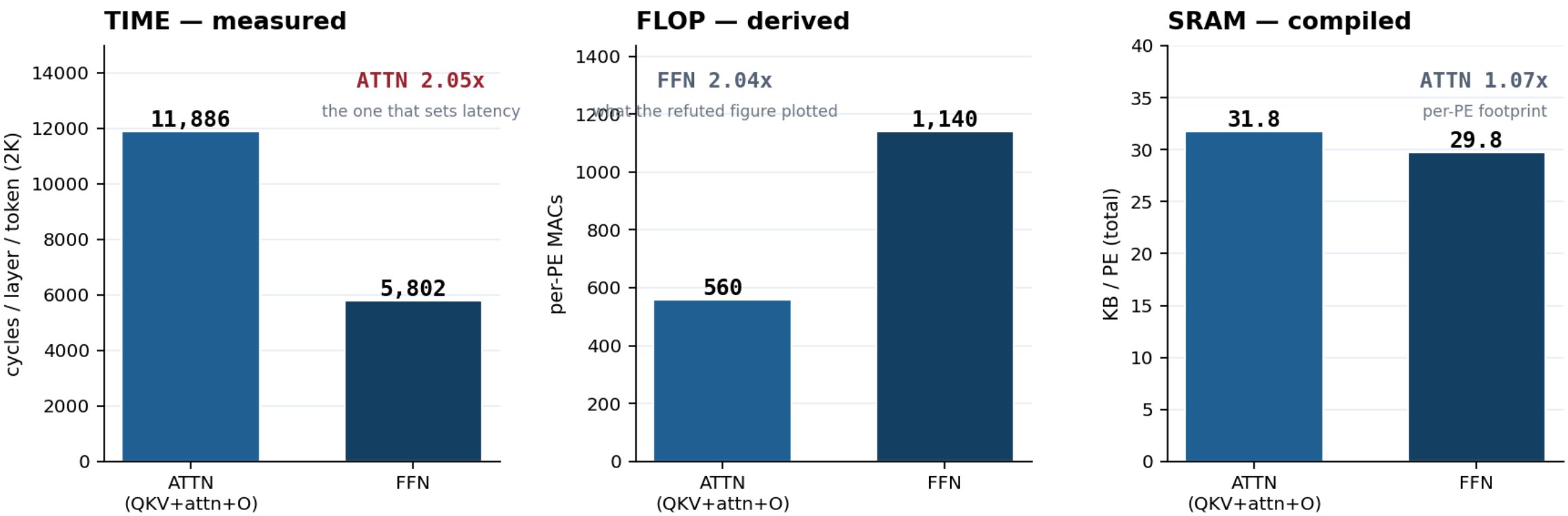
3.7–3.9× the original's single-layer latency in half the PEs

Raw CS-3 device cycles, batch 1, BF16, one layer. Serialized latency, not pipelined throughput; synthetic weights.

INITIAL KV	ORIGINAL ATTN + FFN (131,072 PE)	LAYOUT C A1 + A2 + A3 (65,536 PE)	RATIO
1,280	26,377 cycles	102,511 cycles	3.886×
5,120	28,000 cycles	104,330 cycles	3.726×

FLOP does not predict time here: FFN has 2× the MACs, ATTN takes 2× the time

Original ATTN vs FFN: three axes, three answers — FLOP does NOT predict time here



The original ATTN and FFN blocks on three axes. TIME is 4b-wide-layer's phase-TSC profile (per layer per token, 2K, full 256 × 256 square); FLOP is derived MACs per PE; SRAM is the compiled per-PE total. ATTN also grows with context ($T_{\text{attn}} / T_{\text{fn}} \rightarrow \sim 2.7$ at 8K); FFN is flat.

The kernel is comm- and latency-bound, so a three-way cut pays three coordination costs

- The MAC model that named A3 (FFN) as the pacing stage predicted the ratio and got the mechanism backwards — withdrawn
- Measured: ATTN paces the original at $2.05\times$ FFN's time, and FFN idles
- Per-stage time for A1 / A2 / A3 was never measured; only the combined single-layer numbers exist
- Separately measured: ATTN and FFN as independent pipelined stages give $1.47\times$ at 2K and $1.40\times$ at 8K
- That result keeps two blocks and overlaps them — the lever layout C gave up

The first five-layer group with on-wafer feedback runs on CS-3, exact

- Mini geometry: 5 layers × 33 tokens, 10,560 output elements exact
- Full geometry: 5 layers × 2 tokens, 5,120 exact
- Full cache probe: 83,886,080 K/V values and every cursor pass; a one-bit-flip negative control fails as it should
- Router-only strip transport with the marker / ACK / reset / START protocol validated both directions, with dropped- and duplicated-fragment controls
- Two device defects found and fixed on 09-10: A2 RoPE reset overwriting XKCache; cut-chain zero local-layer count

Closed. Next: one fused 256×256 block with attention state on one PE in N

- The two-group D17 link is paused (Le, 09-13)
- Fused block: FFN + QKV on every PE; attention heavy state on 1-in-N PEs; the rest hold KV and a minimal scan
- Gate: compile-only SRAM of the three PE images
- First-order expectation: $\sim 1.0\times$ the original's latency in half the area — derived, not measured
- Layout C's structural cause — three serial stages sharing one block's area — does not apply to a fused block

06

Why the hardware advantage does not come out

A property of this kernel, not of the architecture — the scoping caveat is on the slide.

87% of per-layer time is independent of how much work each PE does

- Least-squares on the measured height ladder: $T = 10,398 + 855 \times W$ (W = per-PE work) — halving the area moved time by 1.4%
- Per PE per layer: 3,080 bytes in ~17,688 cycles = 0.174 B/cycle against ~27 available, 0.63% utilisation
- Mechanism: 10 hidden + 16 attention values per PE, so a QKV matvec is 160 MACs against ~100 cycles of DSD setup; 8–9.6K of the fixed term is collectives
- Aggregate: 6,269× the bandwidth converted into 4.7×; 0.045% utilisation against the H100's 59.7%
- Scope: Cerebras quotes CS-5 at 10,000 tok/s per user on 31B–120B models — a model 8–30× larger running 10× faster. These are this kernel's numbers, not the architecture's
- Their own rates back-solve to 0.24–1.52% of aggregate bandwidth, so the diagnosis (latency-bound) is corroborated; the lever it implies is speculative decoding, not yet pulled

07

Industry context and positioning

What NVIDIA's LPX and Cerebras's CS-6 say about the boundary, and how to frame the argument.

Both LPX and CS-6 add a DRAM tier; the field is moving from total residency to tiering

- NVIDIA Groq 3 LPX: 500 MB compiler-managed SRAM per chip, 128 GB rack-wide, 96 C2C links = 2.5 TB/s per chip, plus 256 GB + 128 GB DRAM
- LPX does FFN / MoE; the GPUs hold prefill and decode attention and the whole KV — attention—FFN disaggregation with compile-time static allocation
- Shared design constant: a full SRAM sweep is $\sim 3.3 \mu\text{s}$ on LPX and $\sim 2.1 \mu\text{s}$ on WSE-3
- The difference is the boundary: LPX made the chip edge part of the compiler's schedule; Cerebras has no edge inside a wafer
- Cerebras CS-6 (Hot Chips 2026) adds 3D-stacked DRAM — the tier CS-3 lacks. Roadmap, no capacity or bandwidth numbers published
- If KV moved there the thresholds are 4.0 TB/s at 29K, 27.1 TB/s at 197K ($8.1\times$ HBM), 84.6 TB/s at p90 — derived

From capacity to residency

On-chip capacity → performance

- 167K far pool < 197K median
- Numbers are this kernel's, not the architecture's
- CS-6 and LPX both add DRAM

The residency frontier

- Which state must be resident
- Weights tierable; KV not (53×)
- Per-PE distribution binds, not totals

Taxonomy is AFD's table; placement is the measured, novel part; method is what survived this week

- Taxonomy: dense weights tierable; MoE experts must be tiered; KV not tierable (53× short); activations trivially tierable
- Placement: inside the resident tier, per-PE distribution binds, not aggregate capacity — halving width halves the ceiling, fusion doubles it, at identical totals
- Per-PE footprint is not weight-dominated: 24.8 of the ATTN block's 31.8 KB is code, buffers and KV machinery
- Method: across ~12 contested claims this week, the single predictor of survival was whether the measurement carried a control; the failure mode recurred twelve times
- CS-3 is the extremal no-tier point, so every residency decision is forced into the open and measurable

DECIDE

What the room needs to settle

Convention	Clock for every tok/s: 750 MHz as reported, or 0.85 GHz per the 09-07 rule
Fused block	Go / no-go criteria for the compile-only SRAM gate
Workstream	Prefix reuse inside today's 29K, with near-pool harvesting as the mechanism
Measure	The width axis — four points on height, none on width
Measure	Wafer-edge round trip to a GPU — waits on Cerebras opening the API
Framing	Adopt the residency frontier, or keep capacity

C

Corrections

Three things said earlier — one last week, two this week — that the measurements overturned.

The per-PE MAC breakdown is withdrawn: it predicted the 3.7× and named the pacing stage backwards

- As sent: original ATTN 560 / FFN 1,140 MACs per PE; layout C A1 / A2 / A3 = 1,536 / 640 / 3,648, 'explaining 88–92% of the ratio'
- Measured (4b-wide-layer phase-TSC): ATTN takes 2.05× FFN's time — the kernel is comm- and latency-bound, not FLOP-bound
- Withdrawn with it: the per-PE-work bar chart, the 2.65× three-way floor, the 0.31× / 0.62× throughput figures, and 'enlarge FFN to 512 × 256'
- The 3.7–3.9× measurement itself stands; struck sections are kept struck in the conclusion doc so they are not re-derived

The 171 μ s crossing was the public API, not the fabric

- Le: the card-to-card interconnect is 3–5 μ s at 1.2 Tbit/s (150 GB/s); the public Cerebras API does not expose it
- The 171 μ s was measured through that API — 43 $\times$ worse than the silicon
- Withdrawn: Rounds 157–158's 'the boundary is a wall, AFD is dead at bs = 1'
- Survives: KV cannot cross a link (53 $\times$ short); the link carries activations, so AFD is viable — on the weights side
- Revived: PD-split, which had been ruled out on a 3.2 GB/s host-to-wafer measurement of a different leg

The 'median context 309K' was the main-thread number; over all calls it is 197K

- Subagent calls are 44% of the Claude Code trace and were not in the 09-04 / 09-05 cut
- Main thread p50 302K; subagents p50 100K; all calls p50 197K
- Every capacity and spectrum figure this week uses 197K
- The verdict does not change: 197K is still 6.8× the measured ceiling and above the far pool

A

Appendix

The full tables behind the talk, one per slide.

Claude Code per call: p10 / p50 / p90 / p99 / max, tokens

TRACE	N	INPUT CONTEXT	OUTPUT
Claude Code · main thread	26,144	95K / 302K / 719K / 955K / 1.0M	166 / 763 / 3.8K / 10.6K / 64K
Claude Code · subagents	20,506	29K / 100K / 326K / 726K / 991K	2 / 4 / 170 / 1.5K / 22.8K
Claude Code · all calls	46,650	47K / 197K / 616K / 930K / 1.0M	2 / 243 / 2.5K / 8.7K / 64K

Mooncake per request: p10 / p50 / p90 / p99 / max, tokens

Output is hard-capped at 2,000.

TRACE	N	INPUT CONTEXT	OUTPUT
Mooncake · conversation	12,031	964 / 6.9K / 27K / 85K / 126K	24 / 350 / 597 / 1.1K / 2,000 (cap)
Mooncake · toolagent	23,608	1.8K / 6.3K / 17K / 62K / 126K	3 / 30 / 507 / 897 / 2,000 (cap)
Mooncake · synthetic	3,993	30 / 11.6K / 39K / 66K / 191K	4 / 69 / 389 / 768 / 893

ServeGen per request: p50 / p90 / p99 / max, tokens

ServeGen publishes mean / p50 / p90 / p99 / max only — no p10, no raw rows.

MODEL	INPUT CONTEXT	OUTPUT
ServeGen · DeepSeek-R1	208 / 2.1K / 23K / 57K	792 / 3.4K / 12.6K / 32.8K
ServeGen · Qwen-Max	812 / 2.4K / 6.2K / 31K	56 / 433 / 840 / 9.3K
ServeGen · Qwen-Plus	216 / 1.7K / 4.7K / 372K	39 / 213 / 1.1K / 16K
ServeGen · Qwen-Turbo	680 / 813 / 2.3K / 229K	11 / 73 / 467 / 10.9K

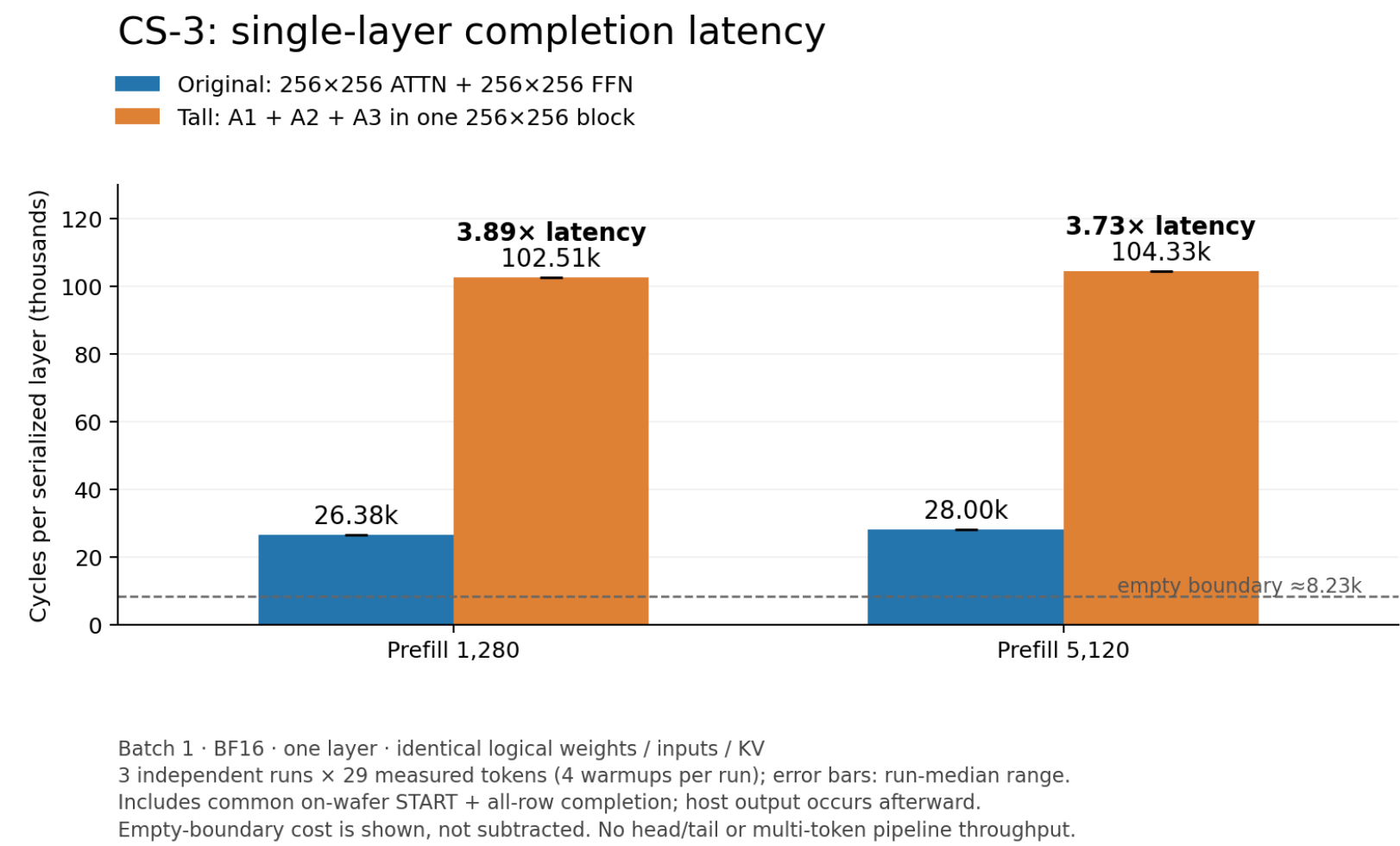
Main thread vs subagents: behaviour

	MAIN THREAD	SUBAGENTS
Output share of the class's token traffic	11.6%	1.5%
Intra-turn gap p50	14.9 s	8.3 s
Calls per session p50 / p90	21 / 267	15 / 46
Output p50 by model	Opus 4.8 837 · Opus 5 673 · Fable 5 501 · Fable 5.1 1,145	Opus 4.8 3 · Opus 5 3 · Sonnet 5 3 · Haiku 4.5 1 (p90 4) · Fable 5 74

Three axes for the original blocks at 2K

BLOCK	PER-PE MACS (DERIVED)	PER-LAYER TIME (MEASURED CS-3)	PER-PE SRAM (COMPILED)
Original ATTN (QKV + attn + O)	560	11,886 cycles	31.75 KB
Original FFN	1,140	5,802 cycles	29.77 KB

Single-layer completion latency, original vs layout C



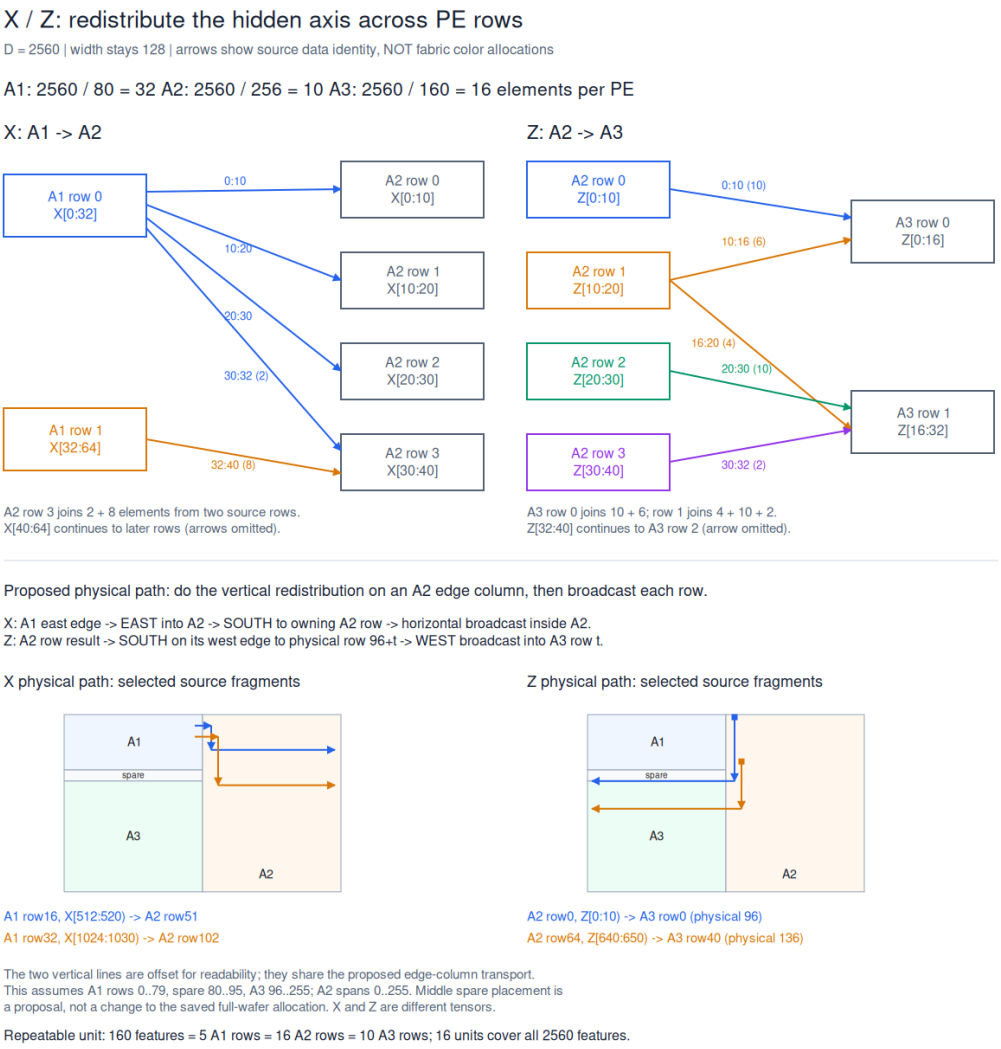
Cycles per serialized layer at two initial KV lengths. Three independent runs × 29 measured tokens; error bars are the run-median range. The dashed line is the empty-boundary cost ($\approx 8.23K$), shown but not subtracted.

One group on a 256 × 256 block

Columns × rows; source of truth is the implementation handoff §3.

STAGE	COLUMNS × ROWS	VALUES PER ROW	PES
A1 (QKV)	128 × 80	hidden 32	10,240
Spare	128 × 16	—	2,048
A3 (FFN)	128 × 160	16	20,480
A2 (attention + O)	128 × 256	10	32,768

Why the row shards are 32 / 10 / 16



How the hidden dimension is re-sharded between stages.

Which tensor axis each stage shards

Weight ownership: change PE-Y hidden slices; keep PE-X shards

Tensor-axis ownership diagram | example local tiles for hidden=2560, width=128 | no matrix transpose proposed



What gets re-sliced along hidden:

- A1: input X, attention RMSNorm weights, and the hidden INPUT dimension of Q/K/V weights.
- A2: residual X/Z and the hidden OUTPUT dimension of O weights.
- A3: FFN RMSNorm weights, UP/GATE hidden INPUT, and DOWN hidden OUTPUT.

Q/K head-norm and RoPE are attached to the unchanged attention-column mapping, not hidden-row slicing.
KV sequence positions use their own row/cursor mapping; hidden re-slicing does not move cached positions.
Grid shows selected rows/columns only. Colors mark row slices; they do not allocate fabric colors.

Weight axes per stage.

Bandwidth a DRAM tier would need to hold the KV scan at today's step time

Derived from the whole-KV-per-token read; CS-6 has published no numbers.

CONTEXT	THRESHOLD	VS H100 HBM
29K	4.0 TB/s	1.2×
197K (Claude Code median)	27.1 TB/s	8.1×
616K (Claude Code p90)	84.6 TB/s	25×

LPX as published

From the NVIDIA blog, as fetched by 4b-wide-layer.

QUANTITY	VALUE
SRAM per chip (compiler-managed)	500 MB; 128 GB rack-wide
On-chip bandwidth	150 TB/s
Chip-to-chip	96 C2C links = 2.5 TB/s per chip; plesiosynchronous
Role in the system	FFN / MoE; GPUs hold attention and the whole KV (AFD)
DRAM	256 GB + 128 GB